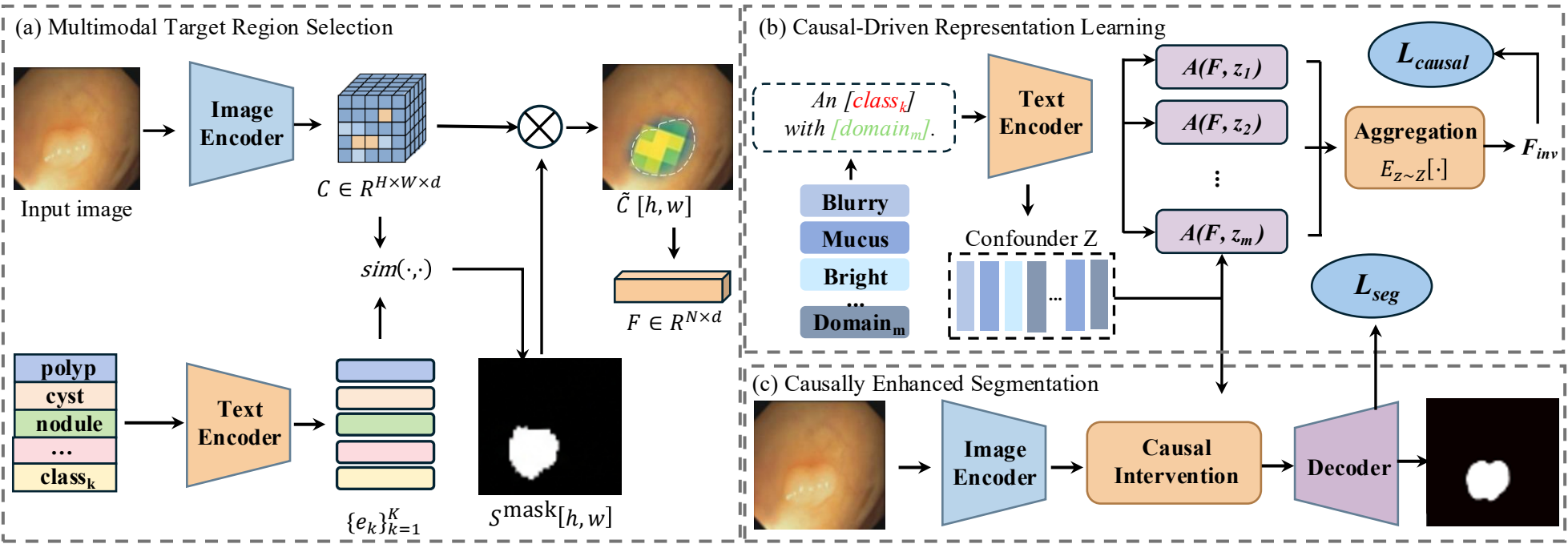
(a) Multimodal Target Region Selection
Input image
Image Encoder
$C \in R^{H \times W \times d}$
$\tilde{C}\ [h, w]$
$F \in R^{N \times d}$
$sim(\cdot,\cdot)$
polyp
cyst
nodule
...
class$_k$
Text Encoder
$\{e_k\}_{k=1}^{K}$
$S^{\text{mask}}[h, w]$
(b) Causal-Driven Representation Learning
An [class$_k$] with [domain$_m$].
Blurry
Mucus
Bright
...
Domain$_m$
Text Encoder
Confounder Z
$A(F, z_1)$
$A(F, z_2)$
$A(F, z_m)$
Aggregation
$E_{z \sim Z}[\cdot]$
F_{inv}
L_{causal}
L_{seg}
(c) Causally Enhanced Segmentation
Image Encoder
Causal Intervention
Decoder